| | | |
|---|---|---|
| $\rho_b$ | Bulk density | M L$^{-3}$ |
| $\rho_s$ | Particle density | M L$^{-3}$ |
| $\rho_w$ | Density of water | M L$^{-3}$ |
| $\sigma$ | Surface tension | M L T$^{-2}$ L$^{-2}$ |
| $\phi$ | Porosity | L$^3$ L$^{-3}$ |
| $\phi_a$ | Air-filled porosity | L$^3$ L$^{-3}$ |

## Problem 2-1

A moist sample of a sandy soil has a field-wet mass ($M_f$) of 1170 g and a volume ($V_t$) of 640 cm$^3$. After oven-drying at 105 °C the dry mass ($M_s$) is determined as 928 g. Assume a typical particle density ($\rho_s$) for mineral soils and calculate:

a. The bulk density ($\rho_b$)

Answer:

$$\rho_b = \frac{M_s}{V_t} = \frac{928}{640} = 1.45 \text{ g cm}^{-3}$$

b. The porosity ($\phi$), assuming a typical particle density for a mineral soil of 2.65 g cm$^{-3}$

Answer:

$$\phi = 1 - \frac{\rho_b}{\rho_s} = 1 - \frac{1.45}{2.65} = 0.45 \text{ cm}^3 \text{ cm}^{-3}$$

c. The gravimetric water content ($\theta_m$)

Answer:

$$\theta_m = \frac{M_f - M_s}{M_s} = \frac{1170 - 928}{928} = 0.26 \text{ g g}^{-1}$$

d. The volumetric water content ($\theta$)

Answer:

$$\theta = \frac{V_w}{V_t} = \frac{M_f - M_s}{\rho_w V_t} = \frac{1170 - 928}{(1)(640)} = 0.38 \text{ cm}^3 \text{ cm}^{-3}$$

Alternatively, the volumetric water content ($\theta$) can be calculated from the soil bulk density and density of water, as well as the gravimetric water content ($\theta_m$):

$$\theta = \theta_m \frac{\rho_b}{\rho_w} = 0.26\frac{1.45}{1} = 0.38 \text{ cm}^3 \text{ cm}^{-3}$$

e. The degree of saturation ($S$).

Answer:

$$S = \frac{\theta}{\phi} = \frac{0.38}{0.45} = 0.84$$

f. The air-filled porosity ($\phi_a$).

Answer:

$$\phi_a = \phi - \theta = 0.45 - 0.38 = 0.07 \text{ cm}^3 \text{ cm}^{-3}$$

## Problem 2-2

A sandy soil with a gravimetric water content ($\theta_m$) of 0.1 kg kg$^{-1}$ was collected for a laboratory study.

a. Determine the mass of soil required to pack a cylindrical soil column with a radius ($r$) of 0.2 m and a height ($h$) of 0.4 m to a bulk density ($\rho_b$) of 1600 kg m$^{-3}$.

Answer:
First, determine the volume ($V_t$) of the cylindrical soil column:

$$V_t = \pi r^2 h = \pi 0.2^2 (0.4) = 0.05 \text{ m}^3$$

Next, calculate the mass of oven dry soil ($M_s$):

$$M_s = \rho_b V_t = 1600(0.05) = 80 \text{ kg}$$

From the dry soil mass and gravimetric water content we can determine the required mass of field-wet soil ($M_f$):

$$\theta_m = \frac{M_f - M_s}{M_s} \rightarrow M_f = M_s(1+\theta_m) = 80(1+0.1) = 88 \text{ kg}$$

b. Calculate the porosity ($\phi$) of the column, assuming a soil particle density ($\rho_s$) of 2650 kg m$^{-3}$.

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Answer:
The porosity may be calculated from bulk density ($\rho_b$) and particle density ($\rho_s$):

$$\phi = 1 - \frac{\rho_b}{\rho_s} = 1 - \frac{1600}{2650} = 0.4 \text{ m}^3 \text{ m}^{-3}$$

c. Determine the volume of water required to fully saturate the column.

Answer:
To determine the volume of water required to saturate the sample we first calculate the volume of the total pore space ($V_p$):

$$V_p = V_t\phi = 0.05(0.4) = 0.02 \text{ m}^3$$

The portion of the total pore space already occupied with water is the mass of water already contained in the soil ($M_w$) divided by the density of water ($\rho_w$) (we assume $\rho_w$ = 1000 kg m$^{-3}$):

$$V_w = \frac{M_f - M_s}{\rho_w} = \frac{88-80}{1000} = 0.008 \text{ m}^3$$

Finally, calculate the volume of water ($V_s$) required to completely saturate the sample:

$$V_s = V_p - V_w = 0.02 - 0.008 = 0.012 \text{ m}^3$$

## Problem 2-3

The side length ($a$) of a weighing lysimeter with a square cross-section is 3 m. The lysimeter was initially filled to a depth ($L$) of 1 m with 11520 kg oven-dry loam soil having a particle density ($\rho_s$) of 2600 kg m$^{-3}$. After a heavy thunderstorm, the total mass of wet soil ($M_f$) in the lysimeter is 13830 kg. After the thunderstorm, an additional 5 cm of water is added by sprinkler irrigation.

a. What is the bulk density of the soil?

Answer:

$$\rho_b = \frac{M_s}{V_t} = \frac{M_s}{a^2 L} = \frac{11520}{9} = 1280 \text{ kg m}^{-3}$$

b. Calculate the average gravimetric ($\theta_m$) and volumetric ($\theta$) water contents of the soil after the end of the thunderstorm (before sprinkler irrigation).

Answer:
The gravimetric water content is:

$$\theta_m = \frac{M_f - M_s}{M_s} = \frac{13830 - 11520}{11520} = 0.2 \text{ kg kg}^{-3}$$

The volumetric water content is:

$$\theta = \frac{V_w}{V_t} = \frac{M_f - M_s}{\rho_w V_t} = \frac{13830 - 11520}{1000(9)} = 0.26 \text{ m}^3 \text{ m}^{-3}$$

c. What is the mass of the soil in the lysimeter and the corresponding volumetric water content after sprinkler irrigation?

Answer:

Final Mass $= M_f + (\text{depth added})\, a^2 \rho_w = 13830 + (0.05)3^2(1000) = 14280$ kg

The final volumetric water content ($\theta$) is

$$\theta = \frac{\text{Final Mass} - M_s}{\rho_w V_t} = \frac{14280 - 11520}{1000(9)} = 0.31 \text{ m}^3 \text{ m}^{-3}$$

## Problem 2-4

A ponded depth ($D$) of 270 mm of water is added to a soil with a volumetric air content of $\theta_a = 0.2$ m$^3$ m$^{-3}$ and a bulk density of $\rho_b = 1450$ kg m$^{-3}$. Calculate the depth of wetting ($L$) assuming complete saturation.

Answer:
We know that the difference between the initial water content and the water content at saturation equals the volumetric air content:

$$\theta_a = \theta_s - \theta_i = 0.2 \text{ m}^3 \text{ m}^{-3}$$

The depth of wetting after addition of 270 mm water is calculated as:

$$L = \frac{D}{\Delta\theta} = \frac{0.27}{0.2} = 1.35 \text{ m}$$

## Problem 2-5

A 2-m deep soil profile is comprised of two layers. The upper layer has a thickness of 0.5 m. The average water contents of the upper and lower layers were determined as $\overline{\theta_1} = 0.21$ and $\overline{\theta_2} = 0.29$ m$^3$ m$^{-3}$.

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Determine the volume of water ($V$) that is stored in the profile within a 1-hectare field.

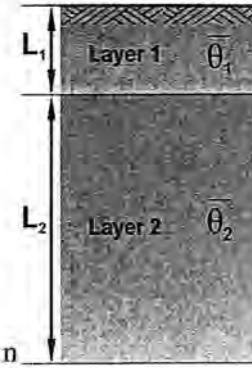


*Fig. 2-1: A layered soil profile*

Answer:
First determine the equivalent depth $D_e$ of water stored in the profile:

$$D_e = \overline{\theta}_1 L_1 + \overline{\theta}_2 L_2 = (0.21)(0.5) + (0.29)(1.5) = 0.54 \text{ m}$$

To obtain the volume, multiply the area (1 ha = $10^4$ $m^2$) with the equivalent depth of water:

$$V = AD_e = 10000(0.54) = 5400 \text{ m}^3$$

## Problem 2-6

A core sample was taken from a 1-m deep soil profile and split into 20-cm increments. After oven drying at 105 °C, gravimetric water content ($\theta_m$) and bulk density ($\rho_b$) were determined for each increment. Then the cores were saturated and the volumetric water contents at field capacity ($\theta_{FC}$) and at the permanent wilting point ($\theta_{WP}$) were determined by means of Tempe cells and the Richards' pressure plate apparatus, respectively.

*Table 2-1: Soil properties*

| **Depth Interval** [cm] | $\theta_m$ [kg kg$^{-1}$] | $\rho_b$ [kg m$^{-3}$] | $\theta_{FC}$ [m$^3$ m$^{-3}$] | $\theta_{WP}$ [m$^3$ m$^{-3}$] |
|---|---|---|---|---|
| 0-20 | 0.09 | 1340 | 0.39 | 0.21 |
| 20-40 | 0.11 | 1425 | 0.36 | 0.23 |
| 40-60 | 0.16 | 1460 | 0.31 | 0.24 |
| 60-80 | 0.19 | 1500 | 0.34 | 0.22 |
| 80-100 | 0.22 | 1510 | 0.33 | 0.24 |

a. Calculate the average volumetric water content and the saturated water content for each depth increment.

Answer:
First we convert the measured gravimetric water contents to volumetric water contents using $\theta = \theta_m \frac{\rho_b}{\rho_w}$ with $\rho_w$ representing the density of water (998.21 kg m$^{-3}$ at 20 °C). We assume that the volumetric saturated water content is equal to porosity with $\rho_s$ as the particle density (2650 kg m$^{-3}$):

$$\phi = \theta_s = 1 - \frac{\rho_b}{\rho_s}$$

The results are listed in the second and third columns of Table 2-2:

b. Estimate the equivalent depth of water needed to completely saturate the profile.

Answer:
For each increment calculate the equivalent depth of water:

$$D_e = L(\theta_s - \theta)$$

as shown in the fourth column of Table 2-2. The results are added to yield 22.6 cm for the 100-cm deep soil profile.

c. Estimate to what depth 15-cm precipitation would saturate the profile.

Answer:
By examining the tabulated results, we observe that 15 cm of precipitation will saturate the first two depth increments (7.4 + 6.0 = 13.4 cm) and that the remaining 15 - 13.4 = 1.6 cm will saturate an additional 1.6/4.4 of the next 20 cm increment. Thus, a reasonable estimate for the wetting depth is 40 + (1.6/4.4) 20 = 47.3 cm.

d. Calculate the plant available soil water in length units within the profile from the volumetric field capacity $\theta_{FC}$ and the wilting point $\theta_{WP}$ listed in Table 2-1.

Answer:
The plant available soil water content for each increment is:

$$D_{PAW} = L(\theta_{FC} - \theta_{WP})$$

with $L$ = 20 cm with the results as shown in the Table 2-2. Adding the values yields a total of 11.8 cm for the total plant available soil water in the profile.

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*Table 2-2: Calculated values*

| Depth Interval [cm] | $\theta$ [cm$^3$ cm$^{-3}$] | $\theta_s$ [cm$^3$ cm$^{-3}$] | $D_e$ [cm] | $D_{PAW}$ [cm] |
|---|---|---|---|---|
| 0-20 | 0.12 | 0.49 | 7.4 | 3.6 |
| 20-40 | 0.16 | 0.46 | 6.0 | 2.6 |
| 40-60 | 0.23 | 0.45 | 4.4 | 1.4 |
| 60-80 | 0.29 | 0.43 | 2.8 | 2.4 |
| 80-100 | 0.33 | 0.43 | 2.0 | 1.8 |
| | | Sum | 22.6 | 11.8 |

## Problem 2-7

A soil has an initial water content of $\theta = 0.1$ m$^3$ m$^{-3}$ and a bulk density of $\rho_b = 1400$ kg m$^{-3}$. What would the final depth of wetting be after internal drainage to field capacity following a rainfall event of 37 mm?

Answer:
First we calculate the volumetric water content at saturation that is equivalent to porosity:

$$\phi = \theta_s = 1 - \frac{\rho_b}{\rho_s} = 1 - \frac{1400}{2650} = 0.47 \text{ m}^3 \text{ m}^{-3}.$$

From the saturated water content we can estimate the water content at field capacity using the "rule of thumb" $\theta_{FC} \cong \frac{\theta_s}{2} = 0.235\,\text{m}^3\,\text{m}^{-3}$.

Once we know the water content at field capacity we can calculate the final depth of wetting:

$$L = \frac{D}{\Delta\theta} = \frac{0.037}{0.235 - 0.1} = 0.27 \text{ m}$$

## Problem 2-8

A neutron probe calibration for a field plot with loamy soil yielded the volumetric water content ($\theta$) - counts per minute ($CPM$) data pairs listed in Table 2-3. The standard reading ($C_s$) during calibration was 19226 counts per minute. The calibration equation is based on the count ratio

(*CR*), which is defined as measured counts per minute divided by the standard count reading: $CR = \frac{CPM}{C_s}$.

*Table 2-3: Calibration data for neutron probe*

| $\theta$ [$m^3\ m^{-3}$] | ***CPM*** | ***CR*** |
|---|---|---|
| 0.07 | 8652 | 0.45 |
| 0.18 | 15381 | 0.80 |
| 0.31 | 24033 | 1.25 |
| 0.36 | 25955 | 1.35 |
| 0.37 | 26916 | 1.40 |
| 0.45 | 30762 | 1.60 |

a. Determine the linear calibration equation that relates volumetric water content in $m^3\ m^{-3}$ (y-axis) to the dimensionless count ratio (x-axis).

Answer:
Use the $\theta$- CR data pairs and perform linear regression analyses with $\theta$ as the dependent variable. We can use spreadsheet software to determine the slope and intercept of the regression line, or perform the calculations manually. The resulting calibration curve is $\theta = -0.0801 + 0.3241CR$, as shown in Fig. 2-2.

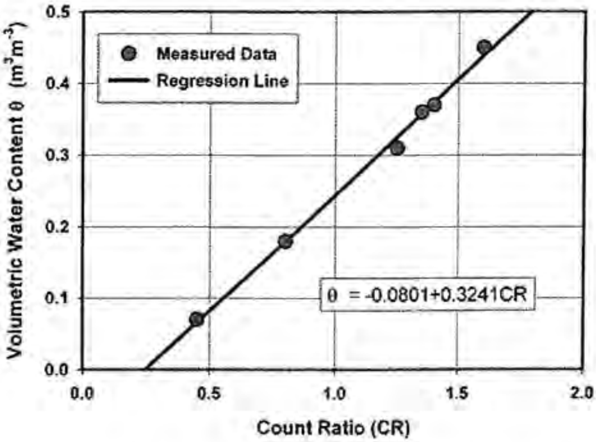


*Fig. 2-2: Neutron probe calibration relationship*

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b. Compute the water content profile measured two weeks later for which the CPM values listed in Table 2-4 were found and for which the standard reading during the measurement was 18720 counts per minute.

Answer:
First we convert the neutron probe counts (*CPM*) to count rate (*CR*) by dividing by the standard reading ($C_s$). Then we apply the calibration equation determined above for each depth, yielding the tabulated and the graphical results below.

*Table 2-4: Water content profile*

| **Depth** [m] | ***CPM*** | ***CR*** | $\theta$ [$m^3 m^{-3}$] |
|---|---|---|---|
| 0.3 | 13206 | 0.71 | 0.149 |
| 0.5 | 13451 | 0.72 | 0.153 |
| 0.7 | 13680 | 0.73 | 0.157 |
| 0.9 | 19455 | 1.04 | 0.257 |
| 1.1 | 19700 | 1.05 | 0.261 |
| 1.3 | 20047 | 1.07 | 0.267 |
| 1.5 | 21110 | 1.13 | 0.285 |
| 1.7 | 21848 | 1.17 | 0.298 |
| 1.9 | 21756 | 1.16 | 0.297 |
| 2.1 | 22386 | 1.20 | 0.308 |

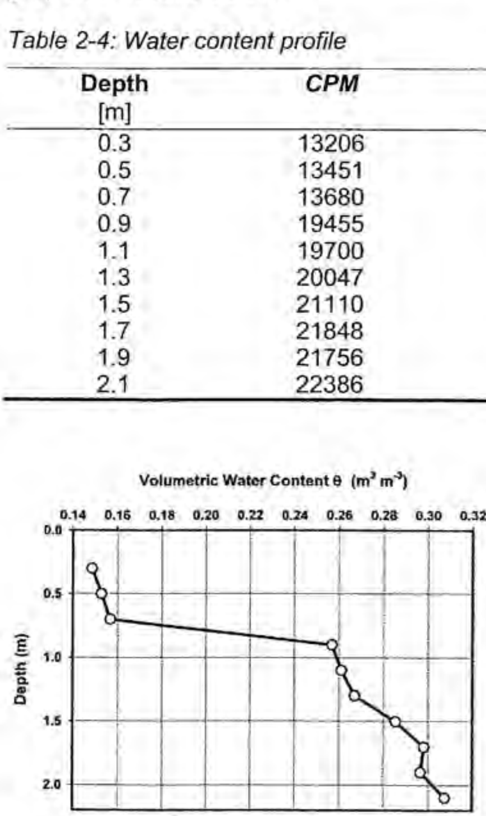


*Fig. 2-3: Water content profile*

## Problem 2-9

The volumetric water content distribution within a soil profile of a cropped field was determined with a neutron probe on two different dates with the following results.

*Table 2-5: Water content distribution within a soil profile on two dates*

| **Depth** [cm] | $\theta$ [m$^3$ m$^{-3}$] Date 1 | Date 2 |
|---|---|---|
| 5 | 0.144 | 0.155 |
| 10 | 0.183 | 0.165 |
| 20 | 0.231 | 0.175 |
| 35 | 0.271 | 0.180 |
| 50 | 0.270 | 0.183 |
| 75 | 0.333 | 0.204 |
| 100 | 0.321 | 0.251 |

a. Assume zero water content on the surface and plot the water content distributions for both days on the same graph.

<u>Answer:</u>

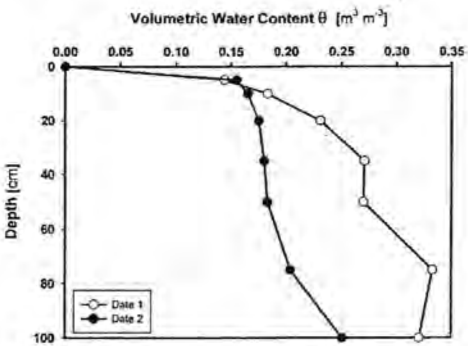


*Fig. 2-4: Water content profiles*

b. Determine the equivalent depth of water stored in each layer and over the entire profile on both dates.

<u>Answer:</u>
Take the average of the water contents between soil layers and multiply values with the layer thickness to find the equivalent depth of water ($D_e$). For the 0 to 5 cm layer for date 1 we calculate:

$$D_e = \frac{0 + 0.144}{2}(5) = 0.36 \text{ m}$$

The equivalent depth of water stored over the entire profile is the sum of the equivalent depths of water of all individual layers.

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*Table 2-6: Equivalent depth of water*

| Depth | $\theta$[cm$^3$ cm$^{-3}$] | | Equivalent Depth $D_e$ [cm] | |
|---|---|---|---|---|
| [cm] | Date 1 | Date 2 | Date 1 | Date 2 |
| 5 | 0.144 | 0.155 | 0.36 | 0.39 |
| 10 | 0.183 | 0.165 | 0.82 | 0.80 |
| 20 | 0.231 | 0.175 | 2.07 | 1.70 |
| 35 | 0.271 | 0.18 | 3.77 | 2.66 |
| 50 | 0.27 | 0.183 | 4.06 | 2.72 |
| 75 | 0.333 | 0.204 | 7.54 | 4.84 |
| 100 | 0.321 | 0.251 | 8.18 | 5.69 |
| | | **Sum** | **26.80** | **18.80** |

c. What is the net change in equivalent depth of water from date 1 to date 2?

Answer:

$$\Delta D_e = D_{e1} - D_{e2} = 26.8 - 18.8 = 8 \text{ cm}$$

d. What might explain the change in the water content over the profile?

Answer:

The measurements show an increase in the water content from Date 1 to Date 2 at the 5-cm depth, which is indicative of surface water application (irrigation or precipitation) between the measurement dates. The overall decrease in the water content deeper in the profile is likely due to drainage and plant transpiration.

## Problem 2-10

A waveguide of length L = 10 cm connected to a Time Domain Reflectometry (TDR) cable tester was successively inserted into a sand, loam, and clay soil. During the measurements, the relative propagation velocity $V_p$ was set at 0.99. The reflected electromagnetic waves were captured from the TDR output screen. The reflection points that mark the entry of the electromagnetic signal into the probe ($x_1$) and the reflection at the end of the probe ($x_2$) are listed together with measured bulk ($\rho_b$) and particle densities ($\rho_s$) in Table 2-7.

*Table 2-7: Soil properties and TDR output for 3 soils*

| | Sand | Loam | Clay |
|---|---|---|---|
| $x_1$ [cm] | 23.639 | 22.989 | 23.076 |
| $x_2$ [cm] | 59.948 | 66.763 | 73.340 |
| $\rho_b$ [kg m$^{-3}$] | 1500 | 1400 | 1200 |
| $\rho_s$ [kg m$^{-3}$] | 2650 | 2650 | 2650 |

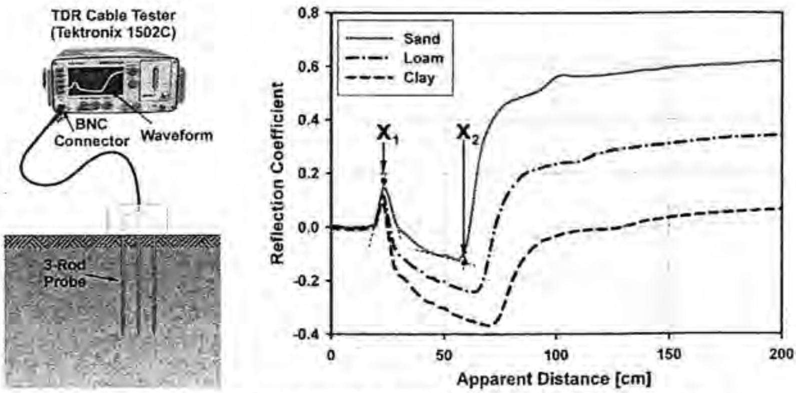


*Fig. 2-5: TDR system and captured reflected waves*

Calculate the bulk dielectric constant $\varepsilon_b$ according to the expression:

$$\varepsilon_b = \left( \frac{x_2 - x_1}{V_p L} \right)^2$$

Answer:
The results are $\varepsilon_b$ = 13.45, 19.55, and 25.78 for sand, loam and clay, respectively.

b. Find the corresponding water contents using Topp's empirical relationship (Topp et al., 1980):

$$\theta = -5.3(10)^{-2} + 2.92(10)^{-2}\varepsilon_b - 5.5(10)^{-4}\varepsilon_b^{\,2} + 4.3(10)^{-6}\varepsilon_b^{\,3}$$

Answer:
The results are $\theta$ = 0.25, 0.34, and 0.41 m$^3$ m$^{-3}$ for sand, loam and clay, respectively.

c. Find the resulting $\theta$ a second time by applying the three-phase mixing model as formulated by Roth et al. (1990):

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$$\varepsilon_b = \left[\theta \varepsilon_w{}^{\beta} + (1-\phi)\varepsilon_s{}^{\beta} + (\phi-\theta)\varepsilon_a{}^{\beta}\right]^{\frac{1}{\beta}}$$

where $\phi$ is the soil's porosity and $\varepsilon_w = 81$, $\varepsilon_s = 4$, and $\varepsilon_a = 1$ are the dielectric constants for water, soil and air, respectively. Take $\beta = 0.5$ which is related to the geometry of the medium in relation to the axial direction of the wave guide.

Answer:

First find the respective porosities ($\phi = 1 - \frac{\rho_b}{\rho_s}$) to be 0.43, 0.47, and 0.55 $m^3\ m^{-3}$ for sand, loam and clay, respectively. Then, use the mixing model to obtain volumetric water contents $\theta$ of 0.26, 0.36, and 0.45 $m^3\ m^{-3}$.

## Problem 2-11

The average TDR-measured soil bulk dielectric constant for silt loam top soil (0 to 0.6 m) having a bulk density ($\rho_b$) of 1325 kg $m^{-3}$ was $\varepsilon_b = 12$. A measurement made with a neutron probe (NP) in a nearby access tube resulted in a count ratio ($CR$) of 0.8 (NP calibration curve: $\theta = 0.02 + 0.45CR$). Calculate the soil water content with both methods, and explain any discrepancies between the methods - state your assumptions!

Answer:

First calculate porosity ($\phi$) by assuming a reasonable value for the particle density ($\rho_s = 2650$ kg $m^{-3}$):

$$\phi = 1 - \frac{\rho_b}{\rho_s} = 1 - \frac{1325}{2650} = 0.5 \text{ m}^3 \text{ m}^{-3}$$

We apply a simplified form of the mixing model used in the previous problem to obtain the volumetric water content from the TDR measurement:

$$\theta = \frac{\sqrt{\varepsilon_b} - (2-\phi)}{8} = \frac{\sqrt{12} - (2-0.5)}{8} = 0.25 \text{ m}^3 \text{ m}^{-3}$$

The water content measured with the neutron probe is calculated using the calibration relationship:

$$\theta = 0.02 + 0.45(CR) = 0.02 + 0.45(0.8) = 0.38 \text{ m}^3 \text{ m}^{-3}$$

The differences might be attributed to differences in soil volumes measured with TDR and the neutron probe, different measurement depths, heterogeneous conditions throughout the profile or bad neutron probe calibration.

## Problem 2-12

The volumetric air content of a soil is 0.15 $m^3 m^{-3}$ and the bulk density is 1280 kg $m^{-3}$. Calculate the volumetric and gravimetric water contents, and determine the soil's bulk dielectric constant.

Answer:
First we calculate the porosity. We assume a particle density of $\rho_s$ = 2650 kg $m^{-3}$.

$$\phi = 1 - \frac{\rho_b}{\rho_s} = 1 - \frac{1280}{2650} = 0.52\,\mathrm{m^3\ m^{-3}}$$

From porosity and known volumetric air content we can calculate the volumetric water content:

$$\theta = \phi - \theta_a = 0.52 - 0.15 = 0.37\ \mathrm{m^3\ m^{-3}}$$

Then we can convert volumetric to gravimetric water content using the relationship:

$$\theta = \theta_m \frac{\rho_b}{\rho_w} \rightarrow \theta_m = \theta \frac{\rho_w}{\rho_b} = 0.37 \frac{1000}{1280} = 0.29\ \mathrm{kg\ kg^{-1}}$$

From known volumetric water content and porosity we can estimate the dielectric constant based on the dielectric mixing model used in the previous 2 problems (Roth et al., 1990):

$$\theta = \frac{\sqrt{\varepsilon_b} - (2-\phi)}{8} \rightarrow \varepsilon_b = (8\theta + 2 - \phi)^2 = (2.96 + 2 - 0.52)^2 = 19.7$$

## Problem 2-13

A clean glass capillary with a radius of 0.01 cm is dipped into a container of water. Determine the water pressure $p_w$ at points A (water-air interface in the capillary), B, and C in Pa, and draw a sketch of the pressure head distribution. Assume that the contact angle between water and clean glass is zero, and the water temperature is 20 °C. The water level (*D*) in the container is 5 cm above the bottom.

Answer:
First we calculate the height of capillary rise ($h_c$):

$$h_c = \frac{2\sigma \cos\gamma}{\rho_w g r} = \frac{(2)(0.0728)(1)}{(998.2)(9.81)(10)^{-4}} = 0.1487\ \mathrm{m}$$

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where $\sigma$ is surface tension of the water-air interface (0.0728 N m$^{-1}$ at 20 °C), $\gamma$ is the water-glass contact angle, $\rho_w$ is the density of water (998.2 kg m$^{-3}$ at 20 °C), $g$ is the acceleration due to gravity (9.81 m s$^{-2}$), and $r$ is the capillary radius.

The pressure in Pa at the water-air interface in point A equals:

$$p_{w-A} = -\rho_w g h_c = -(998.21)(9.81)(0.1487) = -1456 \text{ Pa}$$

The pressure in point B is zero (relative to atmospheric pressure). At point C we have positive hydrostatic pressure equivalent to the vertical distance between point C and the free water level in the container if expressed in length units. The pressure in Pa is given as:

$$p_{w-C} = \rho_w g D = (998.2)(9.81)(0.05) = 490 \text{ Pa}$$

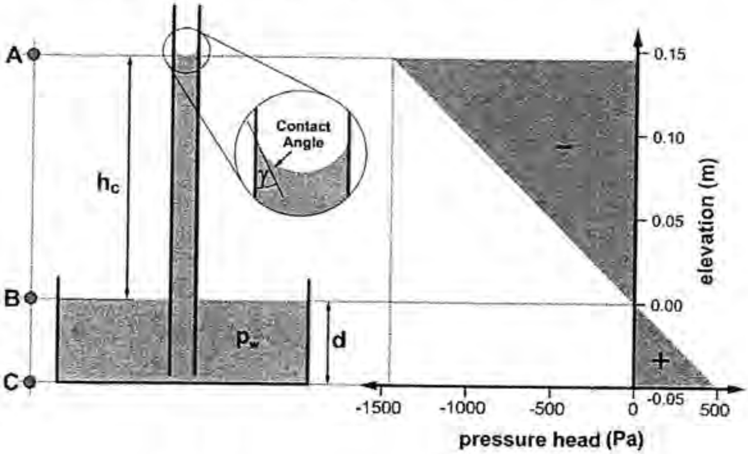


*Fig: 2-6: Capillary rise and pressure distribution*

## Problem 2-14

A small sample of $Ca^{+2}$ montmorillonite is placed in a closed chamber and equilibrated over several days with the chamber environment. The water vapor pressure of the chamber atmosphere after sample equilibration is 514 Pa, the temperature is 20 °C, and the gravimetric water content $\theta_m$ of the sample is 0.15 kg kg$^{-1}$. Assume that the solute (osmotic) potential is negligible and that water is solely retained due to van der Waals interactions in the form of adsorbed films.

a. Calculate the soil water matric potential (expressed first as energy per unit weight and then as energy per unit mass)

Answer:
The Kelvin equation gives the sum of the matric and osmotic potentials. Expressed in units of length, or energy per unit weight:

$$h_m + h_0 = \frac{RT}{M_v g} \ln\left(\frac{e}{e_0}\right)$$

where $h_m$ is the matric potential expressed as a length, $h_0$ is the osmotic potential expressed as a length, $e$ is the water vapor pressure (Pa), $e_0$ is the saturated water vapor pressure at the same temperature, $g$ is the acceleration of gravity (9.81 m s$^{-2}$), $M_v$ is the molecular mass of water (0.018 kg mol$^{-1}$), $R$ is the ideal gas constant (8.31 m$^3$ Pa mol$^{-1}$ K$^{-1}$), and $T$ is absolute temperature (°K).
The saturation vapor pressure at 20 °C (293.15 °K) can be approximated as (Richards, 1971):

$$e_0 = 101325 e^{\left(13.3185 t_R - 1.976 t_R^2 - 0.6445 t_R^3 - 0.1299 t_R^4\right)}$$

with $t_R = 1 - 373.15/T$.
Assuming that $h_o$ is negligible, Kelvin's equation yields:

$$h_m = \frac{(8.31)(293.15)}{(0.018)(9.81)} \ln\left(\frac{514}{2337}\right) = -2.09(10)^4 \text{ m (energy per unit weight)}$$

To express as energy per unit mass, multiply with $g$:

$$E = g h_m = (9.81)(-2.09)(10)^4 = -2.05(10)^5 \text{ J kg}^{-1} \text{ (energy per unit mass)}$$

b. Calculate the thickness of the adsorbed water film.

Answer:
The thickness of the film ($d$) is given in Tuller et al. (1999):

$$d = \sqrt[3]{\frac{A_{svl}}{6\pi\rho_w E}} = \sqrt[3]{\frac{-6(10)^{-20}}{6\pi(998.2)(-205065)}} = 2.496(10)^{-10} \text{ m}$$

where $A_{svl}$ = -6(10)$^{-20}$ J is the Hamaker constant for solid-vapor interactions through the intervening liquid, and $E$ is the disjoining pressure (Derjaguin et al., 1987) expressed in units of energy per unit mass (J kg$^{-1}$).

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c. Calculate the specific surface area ($a_m$) based on the film thickness

Answer:
The mass of water per unit mass of dry soil ($\theta_m$) is equal to the specific surface area ($a_m$) multiplied with film thickness ($d$) multiplied with the density of water ($\rho_w$):

$$\theta_m = a_m d \rho_w$$

The specific surface is:

$$a_m = \frac{\theta_m}{d\rho_w} = \frac{0.15}{2.496(10)^{-10}(998.2)} = 602039 \text{ m}^2 \text{ kg}^{-1} \cong 602 \text{ m}^2 \text{ g}^{-1}$$

## Problem 2-15

A soil sample is in equilibrium with the ambient atmosphere with a relative humidity ($RH$) of 99% and a temperature of 30 °C. The electrical conductivity of the soil solution ($EC_s$) is 25 dS m$^{-1}$.

a. What is the sum of the osmotic and matric potential, expressed as energy per unit weight?

Answer:
The sum of the osmotic and matric potentials is given by the Kelvin equation from the previous problem:

$$h_m + h_0 = \frac{RT}{M_v g}\ln(RH)$$

here $R$ is the ideal gas constant (8.31 Pa m$^3$ mol$^{-1}$ K$^{-1}$), $T$ is absolute temperature in Kelvin, $M_v$ is the molecular weight of water (0.018 kg mol$^{-1}$), $g$ is the acceleration due to gravity (9.81 m s$^{-2}$), and $RH$ represents the ratio of the water vapor pressure of the system to the saturated water vapor pressure (relative humidity).

$$h_m + h_o = \frac{(8.31)(303.15)}{(0.018)(9.81)}\ln(0.99) = -143 \text{ m}$$

b. Determine the osmotic potential in kPa based on its approximate relationship to the electrical conductivity (an approximation because it also depends upon specific ionic composition).

Answer:
Between $EC_s$ = 3 to 30 dS m$^{-1}$ the relation of $EC_s$ and osmotic potential is:

$$E(\text{kPa}) = (-0.36)EC_s(\text{dS m}^{-1})(10)^2 = (-0.36)(25)(10)^2 = -900 \text{ kPa}$$

Converting to Pa and dividing by the product of $\rho_w$ (995.6 kg m$^{-3}$ at 30 °C), and $g$ gives the osmotic potential in units of length ($h_o$):

$$h_o = -\frac{900(10)^3}{(995.6)(9.81)} = -92 \text{ m}$$

c. What is the matric potential?

Answer:
The matric potential is the difference of the last 2 results:

$$h_m = -143 - (-92) = -51 \text{ m}$$

## Problem 2-16

Two tensiometers with a length of 60 cm (measured from the center of the ceramic cup to the vacuum gauge) were installed at a depth of 50 cm. Tensiometer 1 was installed vertically from the soil surface and Tensiometer 2 was installed horizontally from a trench. The matric potential ($h_m$) in vicinity of both tensiometer cups is -1.3 m. What do the vacuum gauges ($h_{gauge}$) of the tensiometers read? Express your results in units of energy per weight (head) as well as in units of energy per volume (soil water pressure).

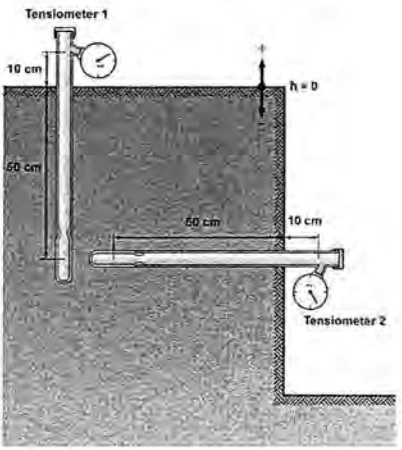


*Fig. 2-7: Tensiometer setup*

Answer:

The pressure head in a tensiometer is reduced by the change in elevation from the cup level to the gauge. The vertical gauge is 60 cm above the level of the cup resulting in

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$h_{gauge1} = -1.3 - 0.6 = -1.9$ m

The horizontal gauge is at the same elevation as the ceramic cup, resulting in

$h_{gauge2} = -1.3 - 0 = -1.3$ m

To express as energy per unit volume or equivalent pressure, multiply with $\rho_w g$ which gives

$$\text{Energy/Volume}_1 = (1000)(9.81)(-1.9) = -(1.86)(10)^4 \text{ Pa} = -18.6 \text{ kPa}$$

and

$$\text{Energy/Volume}_2 = (1000)(9.81)(-1.3) = -(1.28)(10)^4 \text{ Pa} = -12.8 \text{ kPa}$$

## Problem 2-17

A matric potential of $h_m$ = -0.3 m was measured with a tensiometer 0.5 m deep in a uniform soil. Assuming that hydrostatic conditions exist and that the osmotic potential is negligible, determine the distribution of the total hydraulic head ($H$) and pressure head ($h = h_m + h_p$) down to a depth of 2 m. Draw a potential diagram with the gravity, elevation, and total potentials in meters of water on the x-axis, and elevation in meters on the y-axis. If there is a free water table within the profile under consideration, calculate its depth. Designate the soil surface as reference level.

Answer:
The hydraulic head at the 50-cm depth is the pressure head (matric potential here) added to the elevation (-0.5 m relative to the reference level).
Under hydrostatic conditions, $H$ must be constant and the pressure head, expressed in meters, is given by

$H = h - h_z = -0.3 - 0.5 = -0.8$ m

The water table depth $d_{WT}$ is the depth where $h$ is zero

$d_{WT} = 0.8$ m

The plot of the hydraulic head is constant with depth, while the pressure head increases linearly with depth from -0.8 at the surface to zero at the water table. Above the water table it is called matric potential and below the water table it is called pressure potential.

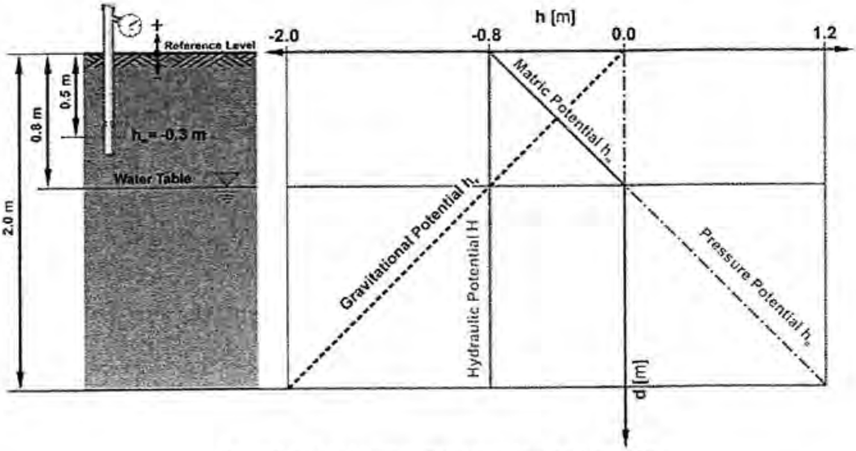


*Fig. 2-8: Potential diagram for soil profile*

## Problem 2-18

Two tensiometers are installed adjacent to each other at 80 cm and 40 cm depths as shown in the sketch below. Which direction does water flow between the two installation depths when the vacuum gauge of Tensiometer 1 reads 19620 Pa and the gauge of Tensiometer 2 reads 20610 Pa? Assume that the osmotic potential is negligible.

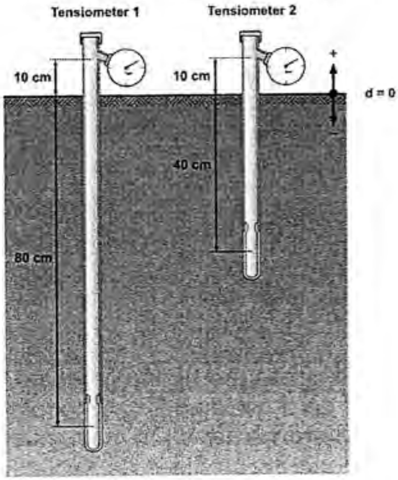


*Fig. 2-9: Tensiometer setup*

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Answer:
A positive value on a vacuum gage indicates the pressure deficit relative to the local atmosphere. Thus, the reading of 19260 corresponds to a pressure of -19260 Pa and 20610 to a pressure of -20610 Pa. To determine the direction of flow we need to calculate the matric potentials at the depths of the tensiometer cups. We first convert the tensiometer readings given in units of energy per volume to pressure head by dividing by $\rho_w g$:

$$h_{gauge1} = \frac{-19620}{(1000)(9.81)} = -2.0 \text{ m}$$

$$h_{gauge2} = \frac{-20610}{(1000)(9.81)} = -2.1 \text{ m}$$

The matric head at each cup is essentially the pressure head, and must be equal to the gauge pressure head added to the decrease in elevation from the gauge to the cup:

$$h_{m1} = -2.0 + 0.9 = -1.1 \text{ m}$$

$$h_{m2} = -2.1 + 0.5 = -1.6 \text{ m}$$

For convenience we choose the soil surface as reference for gravitational and calculate the hydraulic heads $H_1$ and $H_2$ by adding the elevation to the respective pressure head:

$$H_1 = -1.1 + (-0.8) = -1.9 \text{ m}$$

$$H_2 = -1.6 + (-0.4) = -2.0 \text{ m}$$

Since water flow occurs from locations of higher potential energy to locations of lower potential energy in pursuit of equilibrium, the flow direction is upwards (the potential at 40 cm depth is lower than at 80 cm depth).

## Problem 2-19

A static water table exists at the 65 cm depth. Assuming hydrostatic conditions and neglecting osmotic effects, sketch the hydraulic head, the component pressure heads from the soil surface to a depth of 100 cm. Use the bottom of the profile as the reference level.

Answer:
First, we chose a convenient location to calculate the hydraulic head $H$. At the water table, the pressure head ($h$) is zero, yielding $H$ = 35 cm (the elevation ($z$) is defined relative to the reference level). Since, under hydrostatic conditions the hydraulic head along the profile is defined as $H = h + z$, the pressure head is given by $h = 35 - z$.

Note that pressure head ($h$) is used for both positive and negative values relative to atmospheric pressure. A negative $h$ is considered the matric potential, and is represented by $h_m$. A positive $h$ is also referred to as the hydrostatic potential, and is represented by $h_p$.

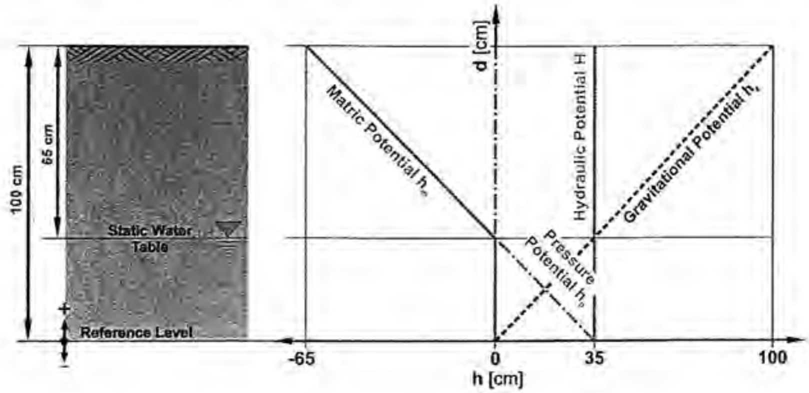


*Fig. 2-10: Distribution of component pressure heads for hydrostatic conditions*

## Problem 2-20

Sketch a typical Soil Water Characteristic (SWC) curve for a sandy and a loam soil. Include both curves in the same graph and plot the volumetric water content on the x-axis and the matric potential on the y-axis. Mark the air entry points and explain the differences in the shapes of the two curves (make sure to correctly label the axes).

Answer:

There are distinct differences between the shapes of the Soil Water Characteristic (SWC) for coarse (sand) and fine textured (loam) soils.

The saturated water content of sand is significantly lower than the saturated water content of loam. While sand has a more homogenous pore-size distribution, where the average pore size is likely larger than loam soil pores, water retention is lower for sand particles. In fact, the graph shows us that more water is retained at saturation by the smaller-diameter loamy soil pores, giving loam larger saturated water content compared to sand.

The air entry value marks the matric potential where the largest pores drain (air enters the soil). Because less energy is required to drain larger

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pores, the air entry potential of the sand is higher (less negative) when compared to the loam.
Finally, there is a distinct difference in the slopes of the two graphs. In general, the finer the soil texture, the greater water retention for a given matric potential, and the more gradual the slope of the curve will be. In sand, most of the water drains close to the air entry potential, as can be seen in the sudden drop in the water content after this point.

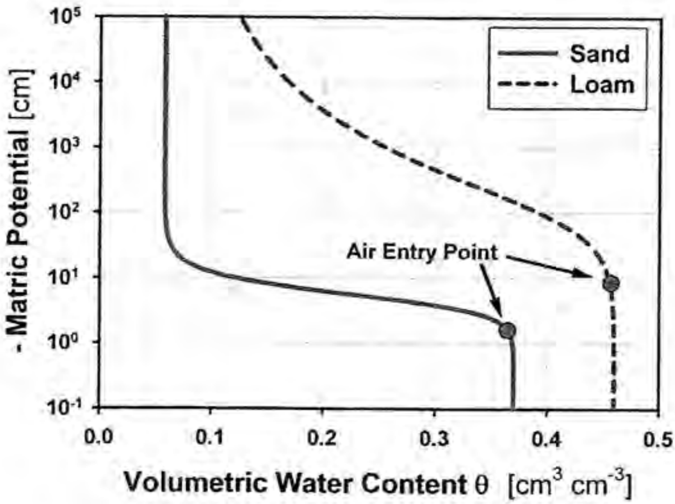


*Fig. 2-11: Soil water characteristic for sand and loam soil*

## Problem 2-21

Calculate and plot the relative saturation ($S_e$) distribution with depth within a bundle of cylindrical quartz capillaries that is vertically dipped into a water reservoir (see sketch). The water temperature is 20 °C and the capillary bundle is comprised of the following capillaries:

*Table 2-8: Size and number of cylindrical capillaries*

| No. of Capillaries | 100 | 300 | 250 | 100 | 18 | 10 | 7 | 4 | 2 |
|---|---|---|---|---|---|---|---|---|---|
| Radius [m] | 1.E-05 | 2.E-05 | 5.E-05 | 1.E-04 | 2.E-04 | 6.E-04 | 8.E-04 | 1.E-03 | 2.E-03 |

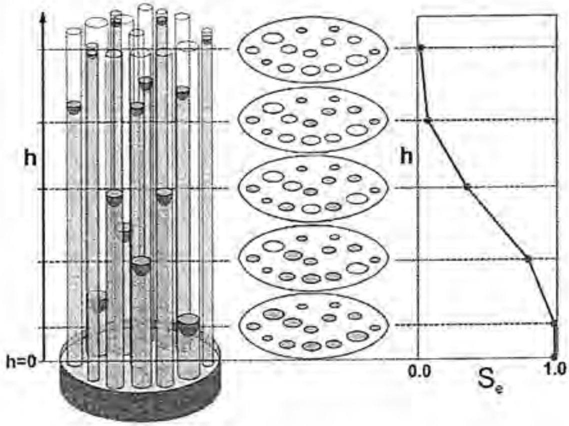


*Fig. 2-12: Bundle of capillaries*

Answer:
First we calculate the height of capillary rise of water (20 °C) for various radii:

$$h_c = \frac{2\sigma \cos\gamma}{\rho_w g r}$$

where $\sigma$ is surface tension of the water-air interface (0.0728 N $m^{-1}$), $\gamma$ is the water-glass contact angle, $\rho_w$ is the density of water (998.2 kg $m^{-3}$), $g$ is the acceleration due to gravity (9.81 m $s^{-2}$), and $r$ is the capillary radius. For each elevation, we then calculate the cross-sectional area of all filled capillaries:

$$A = \pi r^2 n$$

with $n$ as the number of capillaries in each size fraction. We start at the highest elevation where only capillaries with the smallest diameter are filled and then gradually move down to the lowest elevation where all capillaries are filled. At each elevation increment, we add the water filled cross-sectional areas of capillaries with smaller diameters. Finally we calculate the relative saturation at each elevation as the ratio of the water-filled cross-sectional area and the total cross-sectional area of the capillary bundle. For detailed calculations see the related EXCEL file.

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*Table 2-9: Calculation of relative saturation*

| No. of Capillaries | Radius [m] | Capillary Rise [m] | Filled Crossection [m$^2$] | Relative Saturation |
|---|---|---|---|---|
| 100 | 1.E-05 | 1.487 | 3.14E-08 | 4.43E-04 |
| 300 | 2.E-05 | 0.743 | 4.08E-07 | 5.76E-03 |
| 250 | 5.E-05 | 0.297 | 2.37E-06 | 3.35E-02 |
| 100 | 1.E-04 | 0.149 | 5.51E-06 | 7.78E-02 |
| 18 | 2.E-04 | 0.074 | 7.78E-06 | 1.10E-01 |
| 10 | 6.E-04 | 0.025 | 1.91E-05 | 2.69E-01 |
| 7 | 8.E-04 | 0.019 | 3.32E-05 | 4.68E-01 |
| 4 | 1.E-03 | 0.015 | 4.57E-05 | 6.45E-01 |
| 2 | 2.E-03 | 0.007 | 7.09E-05 | 1.00E+00 |

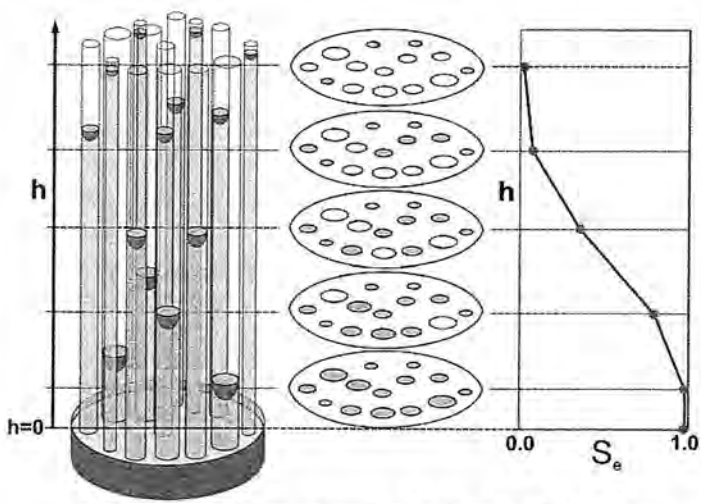


*Fig. 2-13: Relative saturation for bundle of capillaries*

## Problem 2-22

A hanging water column was used to measure part of the desorption branch of a soil water characteristic curve. After equilibration, the vertical distance between the free water level in burette and the bottom of the 5-cm high soil sample is 25 cm (see sketch). Use the top of the soil sample as the reference level and determine the hydraulic, matric, pressure, and gravitational potentials for points A to D extending from the soil into the water tube.

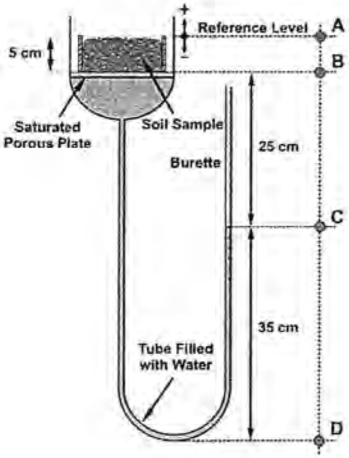


*Fig. 2-14: Hanging water column*

Draw a potential diagram for the system with the potential components on the x-axis and elevation on the y-axis.

Answer:
Expressing potentials as energy per unit of weight and with the sample top as reference, the gravitational potential components are simply the negative vertical distances from the reference level to the points of interest.
The matric potentials in points A and B are -30 and -25 cm (distance between points and the water level in the burette. The pressure potentials are zero (i.e., there are no positive pressures).
In point C (water level) the matric and pressure potentials are zero. In point D the matric potential is zero and the pressure potential is 35 cm. The total hydraulic potential (sum of individual components) is constant at -30 cm throughout the system (equilibrium). We can combine results in tabulated form and draw the potential diagram.

*Table 2-10: Head components of hanging water column*

| Point | $H$ [cm] | $h_z$ [cm] | $h_m$ [cm] | $h_p$ [cm] |
|---|---|---|---|---|
| A | -30 | 0 | -30 | 0 |
| B | -30 | -5 | -25 | 0 |
| C | -30 | -30 | 0 | 0 |
| D | -30 | -65 | 0 | 35 |

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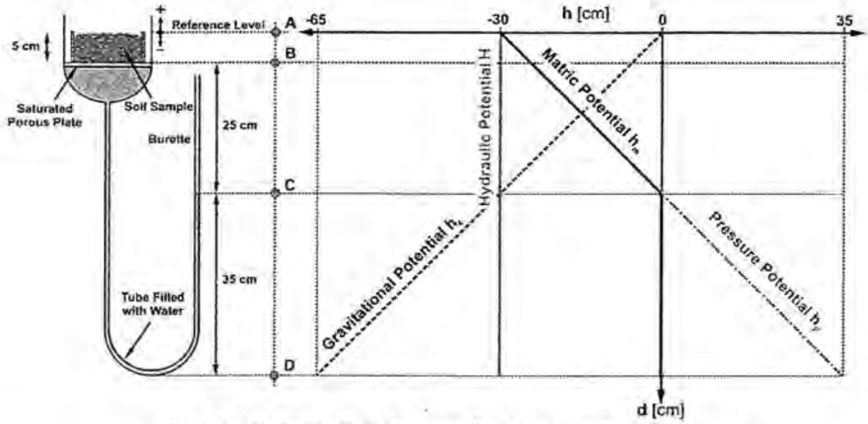



*Fig. 2-15: Potential diagram for hanging water column*

## Problem 2-23

The van Genuchten (VG) model parameters (van Genuchten, 1980) and saturated and residual water contents were determined for a Millville silt loam soil ($\alpha_{VG}$ = 0.394 m$^{-1}$, $n$ = 1.420, $\theta_s$ = 0.5 m$^3$ m$^{-3}$, $\theta_r$ = 0.021 m$^3$ m$^{-3}$). Calculate the equivalent depth of plant available soil water ($D_{PAW}$) stored in the top 1 m of the soil. Use the "Field Capacity - Wilting Point" concept for your calculations. List practical applications that require knowledge of the equivalent depth of plant available soil water.

Answer:

The volumetric water content $\theta_{FC}$ at field capacity ($h$ = -3 m):

$$\theta_{FC} = \theta_r + (\theta_s - \theta_r)\left(\frac{1}{1+(\alpha_{VG}|h|)^n}\right)^m, \quad m = \frac{n-1}{n}$$

$$\theta_{FC} = 0.021 + (0.5 - 0.021)\left(\frac{1}{1+\left[(0.394)(3)\right]^{1.42}}\right)^{0.296} = 0.4 \text{ m}^3 \text{ m}^{-3}$$

Volumetric water content $\theta_{WP}$ at the wilting point ($h$ = -150 m):

$$\theta_{WP} = 0.021 + (0.5 - 0.021)\left(\frac{1}{1+\left[(0.394)(150)\right]^{1.42}}\right)^{0.296} = 0.11 \text{ m}^3 \text{ m}^{-3}$$

Plant available soil water $\theta_{PAW}$:

$$\theta_{PAW} = \theta_{FC} - \theta_{WP} = 0.4 - 0.11 = 0.29 \text{ m}^3 \text{ m}^{-3}$$

Equivalent depth ($D_{PAW}$) of plant available soil water stored in the top 1 m:

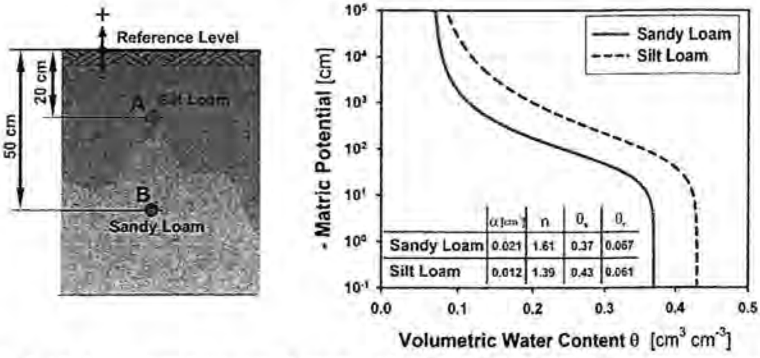
$$D_{PAW} = L\left(\theta_{FC} - \theta_{WP}\right) = 1\left(0.29\right) = 0.29 \text{ m} = 290 \text{ mm}$$

Practical applications that require knowledge of the equivalent plant available soil water include

- Irrigation scheduling to avoid over-irrigation beyond field capacity (leaching losses) or under-irrigation resulting in drought stress.
- To select crops suitable for a certain location, in accordance with available soil and water resources.

## Problem 2-24

Time Domain Reflectometry (TDR) was used to measure the volumetric water contents in points A and B, 20 and 50 cm below the surface of a highly heterogeneous soil profile. Point A lies within a region of silt loam and the soil in point B is located in a sandy loam. The TDR measurements showed identical volumetric water contents of 0.25 $cm^3\,cm^{-3}$ in both depths. Use the Soil Water Characteristic (SWC) information (van Genuchten parameters) provided below for both soil regions to determine the direction of water flow between the two locations. Assume that the osmotic potential is negligible and use the soil surface as reference level.




| | $\alpha$ [$cm^{-1}$] | n | $\theta_s$ | $\theta_r$ |
|---|---|---|---|---|
| Sandy Loam | 0.021 | 1.61 | 0.37 | 0.067 |
| Silt Loam | 0.012 | 1.39 | 0.43 | 0.061 |

*Fig. 2-16: Heterogeneous soil profile and Soil Water Characteristic information*

Answer:

We use the van Genuchten (VG) parametric model for the Soil Water Characteristic (SWC) to relate the measured volumetric water content to

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matric potential. First we rearrange the VG equation to obtain an explicit expression for matric potential $h_m = h$:

$$\theta = \theta_r + (\theta_s - \theta_r)\left(\frac{1}{1+(\alpha_{VG}|h|)^n}\right)^m \quad \Rightarrow \quad h = \frac{1}{\alpha_{VG}}\left[\left(\frac{\theta-\theta_r}{\theta_s-\theta_r}\right)^{-\frac{1}{m}} - 1\right]^{\frac{1}{n}}$$

where $\theta_s$ is the saturated water content, $\theta_r$ the residual water content and $\alpha$, $n$, and $m$ are the VG model parameters. Note that the parameter $m$ is related to $n$ according to the following expression:

$$m = \frac{n-1}{n}$$

For point A we obtain:

$$|h_A| = \frac{1}{0.012}\left[\left(\frac{0.250-0.061}{0.430-0.061}\right)^{-\frac{1}{0.281}} - 1\right]^{\frac{1}{1.390}} = 430.94 \text{ cm}$$

For point B we obtain:

$$|h_B| = \frac{1}{0.021}\left[\left(\frac{0.250-0.067}{0.370-0.067}\right)^{-\frac{1}{0.379}} - 1\right]^{\frac{1}{1.610}} = 89.92 \text{ cm}$$

With the matric potentials determined, we can now calculate the hydraulic potentials ($H$) at both points. Please note that the van Genuchten equation uses the absolute value of the matric potential, but that potential values are actually negative.

For point A we obtain: $H_A = h_A + h_z = -430.94 + (-20) = -450.94$ cm

For point B we obtain: $H_B = h_B + h_z = -89.92 + (-50) = -139.92$ cm

Since water flow occurs from locations of higher potential energy to locations of lower potential energy in pursuit of equilibrium, the flow direction is from point B to point A (upwards).

◆ ◆ ◆

*Logic will get you from A to B. Imagination will take you everywhere – Albert Einstein*